

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
6.	Event Location in Dynamic Phenomena	4	Students will be able to locate the source of rock burst



Event Location

- ✓ Location of bumps, bursts, and related events is important to operations and understanding of the excavation environment.
- ✓ Spatial arrays of geophones serve the purpose.
- ✓ The basic principle is simply (distance)=(velocity)(time).
- ✓ Suppose an event occurs at (x_s, y_s, z_s) at time (t_s) in a rock mass where wave velocity is v .
- ✓ A geophone or receiver “i” senses the wave from the source “s” at time t_i .
- ✓ The distance from source to receiver is traveled by the wave according to the basic principle during the intervening time. Thus,

$$\sqrt{(x_i - x_s)^2 + (y_i - y_s)^2 + (z_i - z_s)^2} = v(t_i - t_s)$$

- ✓ relative to a rectangular coordinate system and master clock that keeps the same time at all receivers.
- ✓ There are four unknowns in the above equation, i.e., (x_s, y_s, z_s, t_s) .
- ✓ However, above equation is not a linear expression, so some thought is required to compute event locations, especially when there are many geophones in an array. In this regard, event location in one, two, and three spatial dimensions is instructive.

Event Location

- ✓ For details one may consult Hardy (2003) where the focus is on laboratory measurements.
- ✓ Applications of microseismic monitoring in underground mining are described by Blake, Leighton, and Duvall (1974).
- ✓ Source and receivers on a line are shown in Figure 9.17.
- ✓ Above Equation has a simple, linear expression in this case. Thus,

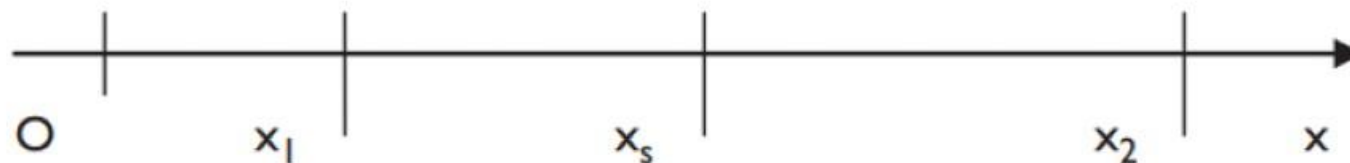


Figure 9.17 Source and receivers on a line with source between receivers.

$$(x_1 - x_s) = v(t_1 - t_s)$$

$$(x_2 - x_s) = v(t_2 - t_s)$$

$$x_s = \frac{(x_1 + x_2)}{2} - \frac{v(t_2 - t_1)}{2} \text{ and}$$

$$t_s = \frac{(t_1 + t_2)}{2} - \frac{(x_2 - x_s)}{2v}$$

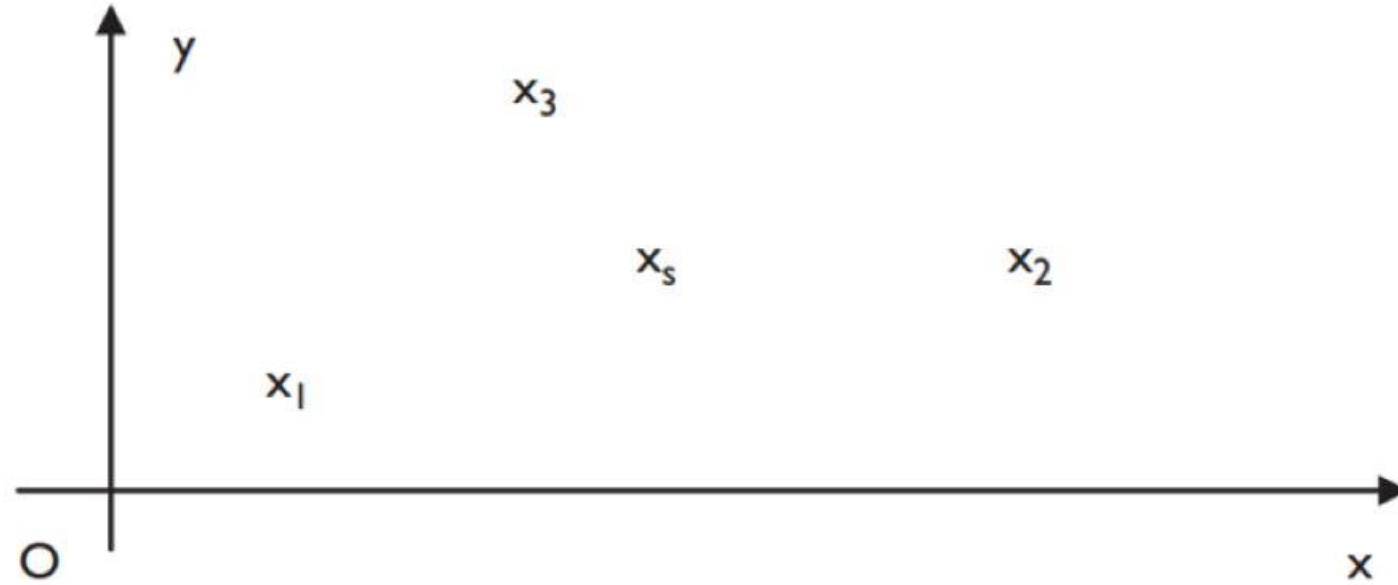


Figure 9.18 Event and receivers in a plane with event source inside the receiver triangle.

- ✓ An event on a plane is shown in relationship to three receivers in Figure 9.18.
- ✓ Application of main equation gives three equations in the three unknowns (x_s, y_s, t_s) . Thus

$$\sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2} = v(t_1 - t_s)$$

$$\sqrt{(x_2 - x_s)^2 + (y_2 - y_s)^2} = v(t_2 - t_s)$$

$$\sqrt{(x_3 - x_s)^2 + (y_3 - y_s)^2} = v(t_3 - t_s)$$



Event Location

- ✓ An event on a plane is shown in relationship to three receivers in Figure 9.18.
- ✓ Application of main equation gives three equations in the three unknowns (x_s, y_s, t_s) . Thus

$$\sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2} = v(t_1 - t_s)$$

$$\sqrt{(x_2 - x_s)^2 + (y_2 - y_s)^2} = v(t_2 - t_s)$$

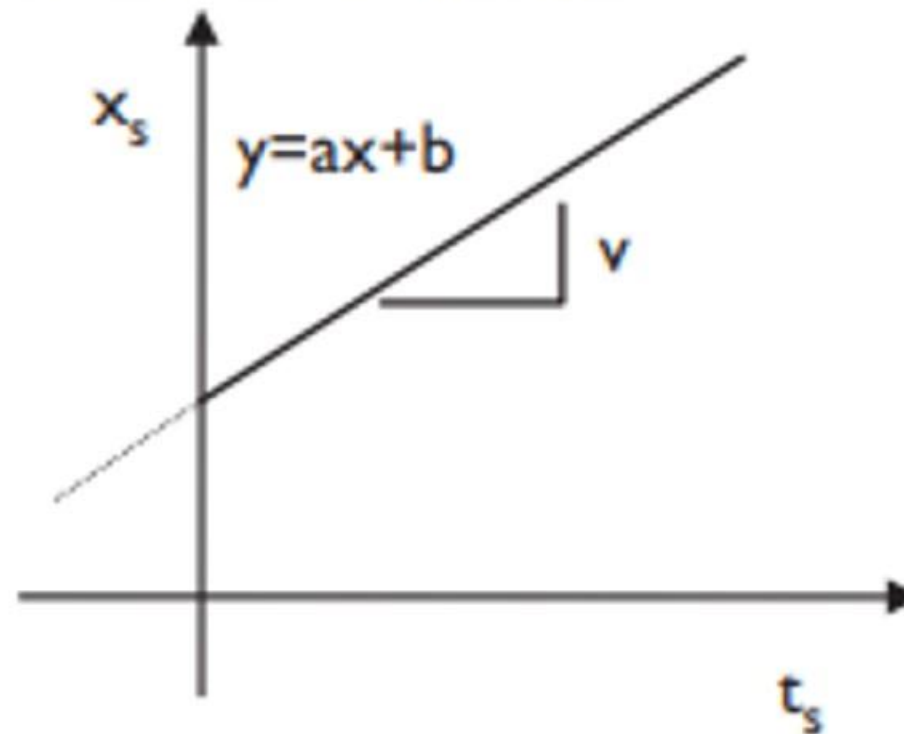
$$\sqrt{(x_3 - x_s)^2 + (y_3 - y_s)^2} = v(t_3 - t_s)$$

- ✓ This system can be reduced to a linear system by squaring each equation in turn and then subtracting a from b, b from c, and c from a.
- ✓ This procedure eliminates the squared terms (x_s^2, y_s^2, t_s^2) with the result



Event Location

- ✓ **Example 9.18** Consider an event source on a line passing through receivers and source. Determine from the basic principle of $(\text{distance}) = (\text{velocity})(\text{time})$, the source location and time of event when locations are ordered such that $x_1 < x_2 < x_s$ and when $x_s < x_1 < x_2$, that is, when the source is on the same side of the receivers rather than in between.



Sketch for Example 9.18

Event Location

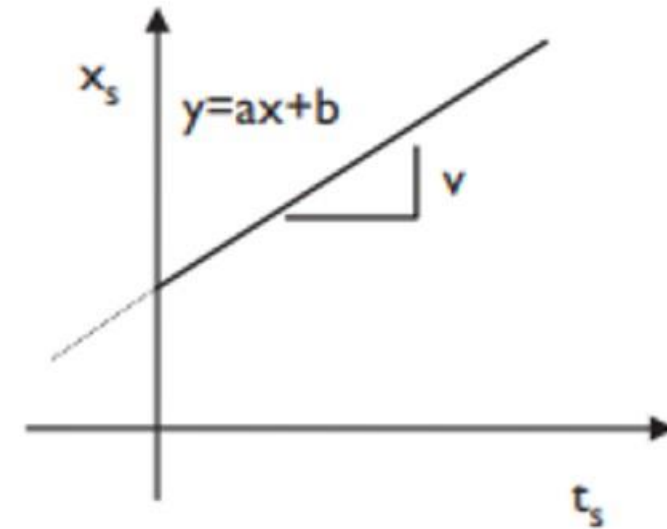
✓ **Example 9.18** Consider an event source on a line passing through receivers and source. Determine from the basic principle of (distance)=(velocity)(time), the source location and time of event when locations are ordered such that $x_1 < x_2 < x_s$ and when $x_s < x_1 < x_2$, that is, when the source is on the same side of the receivers rather than in between.

✓ **Solution:** From the basic distance - time principle

✓ $(x_1 - x_s) = v(t_1 - t_s)$ or $x_s = (x_1 - vt_1) + vt_s$

✓ $(x_2 - x_s) = v(t_2 - t_s)$ or $x_s = (x_2 - vt_2) + vt_s$

✓ These last two equations are straight lines when plotted in an t_s - x_s plane, as shown in the sketch.



Sketch for Example 9.18

✓ In fact the two lines are the same as can be shown by subtracting

✓ the last two equations to obtain $(x_2 - x_1) = v(t_2 - t_1)$

✓ After substitution of this result into the second equation for source location, one has

✓ $x_s = (x_2 - vt_2) + vt_s = (x_1 - vt_1) + vt_s$

✓ Thus, one can determine wave velocity from the observations in this case, but the source location cannot be determined because there is only one independent equation but two unknowns.



- ✓ **Example 9.19** Consider an event on a line such that the event occurs at the location of one of the receivers. Determine whether this event can be located from receiver recordings.

- ✓ **Example 9.19** Consider an event on a line such that the event occurs at the location of one of the receivers. Determine whether this event can be located from receiver recordings.
- ✓ **Solution:** The given data are

$$x_s = x_1 < x_2$$

$$t_s = t_2$$

- ✓ From the basic principle

$$(x_1 - x_s) = v(t_1 - t_s) = 0 \text{ or } (x_2 - x_s) = v(t_2 - t_s) = 0$$

- ✓ In the first case $x_1 = x_s$ and in the second case $x_2 = x_s$.
- ✓ Thus, in both cases the event location is determined.



- ✓ **Example 9.20** Consider a receiver array on a plane as shown in Figure 9.18 and suppose an event occurs at one of the receivers, say, x_2 . Show that the event can be located by receiver data.



Event Location

✓ **Example 9.20** Consider a receiver array on a plane as shown in Figure 9.18 and suppose an event occurs at one of the receivers, say, x_2 . Show that the event can be located by receiver data.

✓ **Solution:** In this case, the basic principle (9.16) shows that

$$\sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2} = v(t_2 - t_s)$$

✓ The arrival time at the receiver $t_2 = t_s$ according to the given conditions. Therefore

$$(x_2 - x_s)^2 = -(y_2 - y_s)^2$$

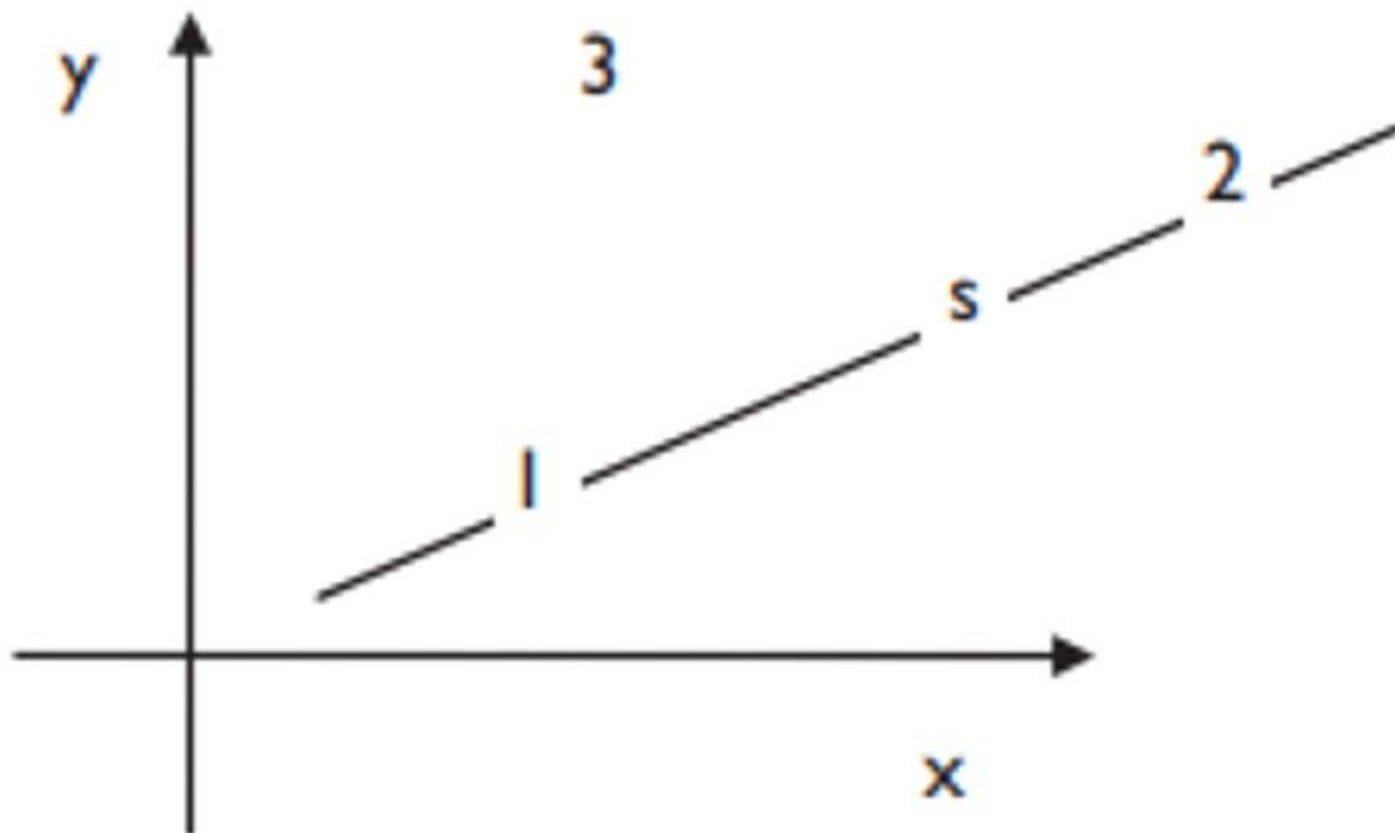
✓ This result requires both sides of the equation to equal zero (all numbers are real).

✓ Thus, $x_2 = x_s$ and $y_2 = y_s$ and the event can be located by the receiver data.



Event Location

- ✓ **Example 9.21** Suppose planar array of receivers detects an event that occurs on a line between two receivers, say, between receivers 1 and 2. Determine whether the event can be located from the receiver data.



Sketch for Example 9.21

Event Location

✓ **Example 9.21** Suppose planar array of receivers detects an event that occurs on a line between two receivers, say, between receivers 1 and 2. Determine whether the event can be located from the receiver data.

✓ **Solution:** The sketch shows the situation.

✓ Application of (9.16) to receivers 1 and 2 gives

$$\sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2} = v(t_1 - t_s)$$

$$\sqrt{(x_2 - x_s)^2 + (y_2 - y_s)^2} = v(t_2 - t_s)$$

✓ Rotation of the axes to bring the new x-axis on line with

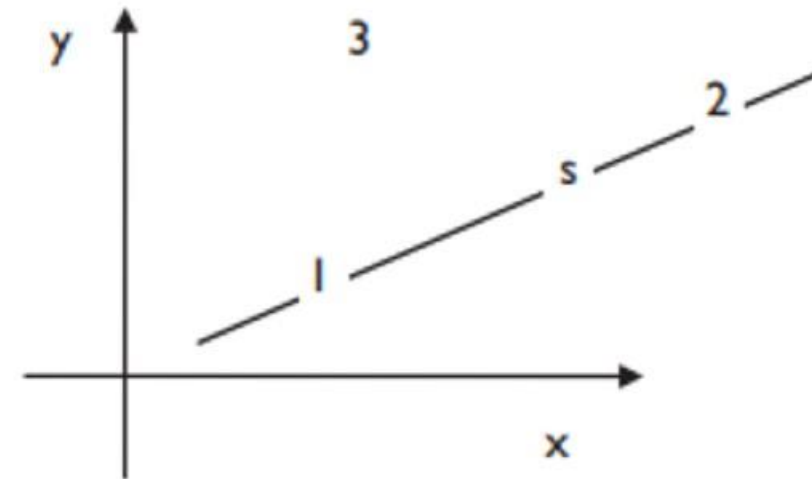
✓ receivers 1 and 2 transforms these two equations into

$$(x_s - x'_1) = v(t_1 - t_s)$$

$$(x'_2 - x_s) = v(t_1 - t_s)$$

✓ with proper choice of the square root operation.

✓ This system is just the system for locating an event between two receivers on a line described in the text. Therefore the event can be located, provided it occurs between receivers.

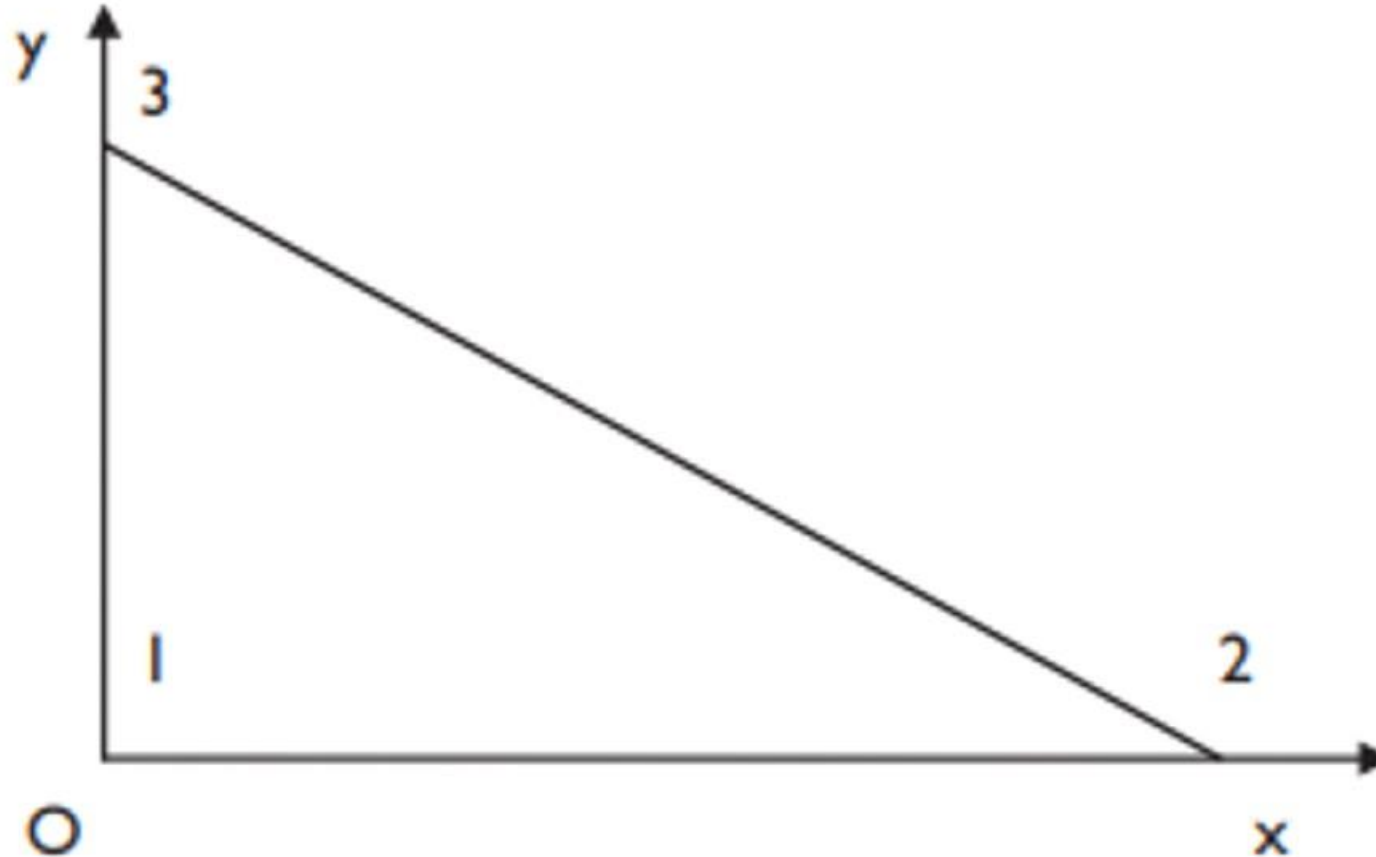


Sketch for Example 9.21



Event Location

- ✓ **Example 9.22** Consider a planar array of three receivers located at the vertices of a right triangle and suppose arrival times are the same at each receiver. Determine the event location and time.



Sketch for Example 9.22

Event Location

✓ **Example 9.22** Consider a planar array of three receivers located at the vertices of a right triangle and suppose arrival times are the same at each receiver. Determine the event location and time.

✓ **Solution:** The sketch shows the receiver array in a plane with origin fixed at the 1-receiver.

$$\sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2} = v(t_1 - t_s)$$

$$\sqrt{(x_2 - x_s)^2 + (y_2 - y_s)^2} = v(t_2 - t_s)$$

$$\sqrt{(x_3 - x_s)^2 + (y_3 - y_s)^2} = v(t_3 - t_s)$$

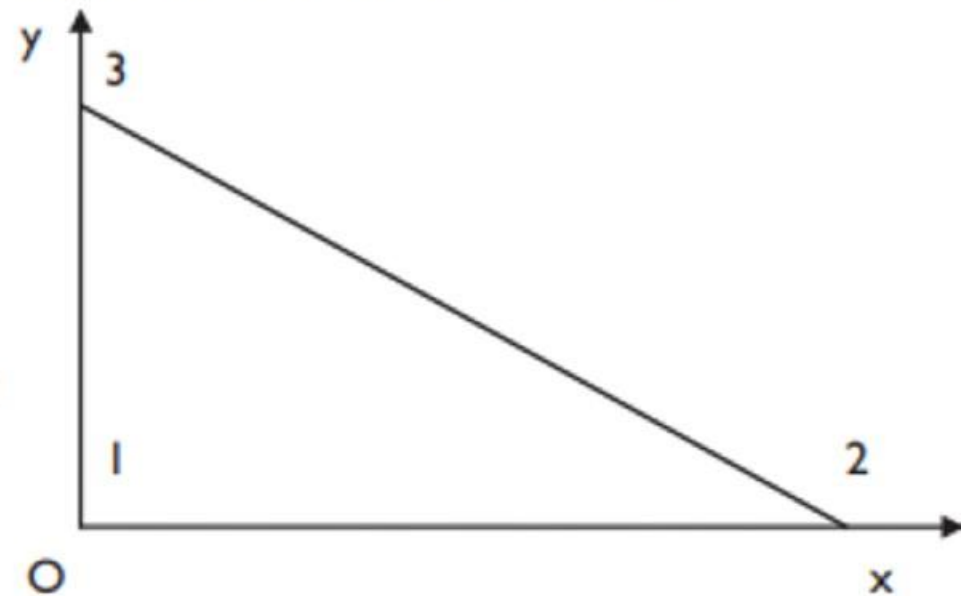
✓ Because the arrival times are the same, the distance between
 ✓ receivers and source must also be the same.

✓ In consideration of the receiver locations then,

$$(-x_s)^2 + (-y_s)^2 = (x_2 - x_s)^2 + (-y_s)^2 = (-x_s)^2 + (y_3 - y_s)^2$$

✓ Solution of the first two equations shows that $x_s = \frac{x_2}{2}$. Similarly, solution of the first and third equations

shows that $y_s = \frac{y_3}{2}$. The event time may be obtained by back substitution. Thus, $t_s = t_1 + \frac{\sqrt{(\frac{x_2}{2})^2 + (\frac{y_3}{2})^2}}{v}$



Sketch for Example 9.22



- ✓ **Example 9.23** Show that when two receivers detect an event at the same time, the source must lie on a perpendicular bisector of the line passing through the two receivers.



✓ **Example 9.23** Show that when two receivers detect an event at the same time, the source must lie on a perpendicular bisector of the line passing through the two receivers.

✓ **Solution:** distance travel - time relationships for the two receivers are

$$\sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2 + (z_1 - z_s)^2} = v(t_1 - t_s)$$

$$\sqrt{(x_2 - x_s)^2 + (y_2 - y_s)^2 + (z_2 - z_s)^2} = v(t_2 - t_s)$$

✓ Rotation of axes to bring the new x-axis on line through the two receivers shows that

$$(-x_s)^2 + (-y_s)^2 + (-z_s)^2 = (x_2 - x_s)^2 + (-y_s)^2 + (-z_s)^2$$

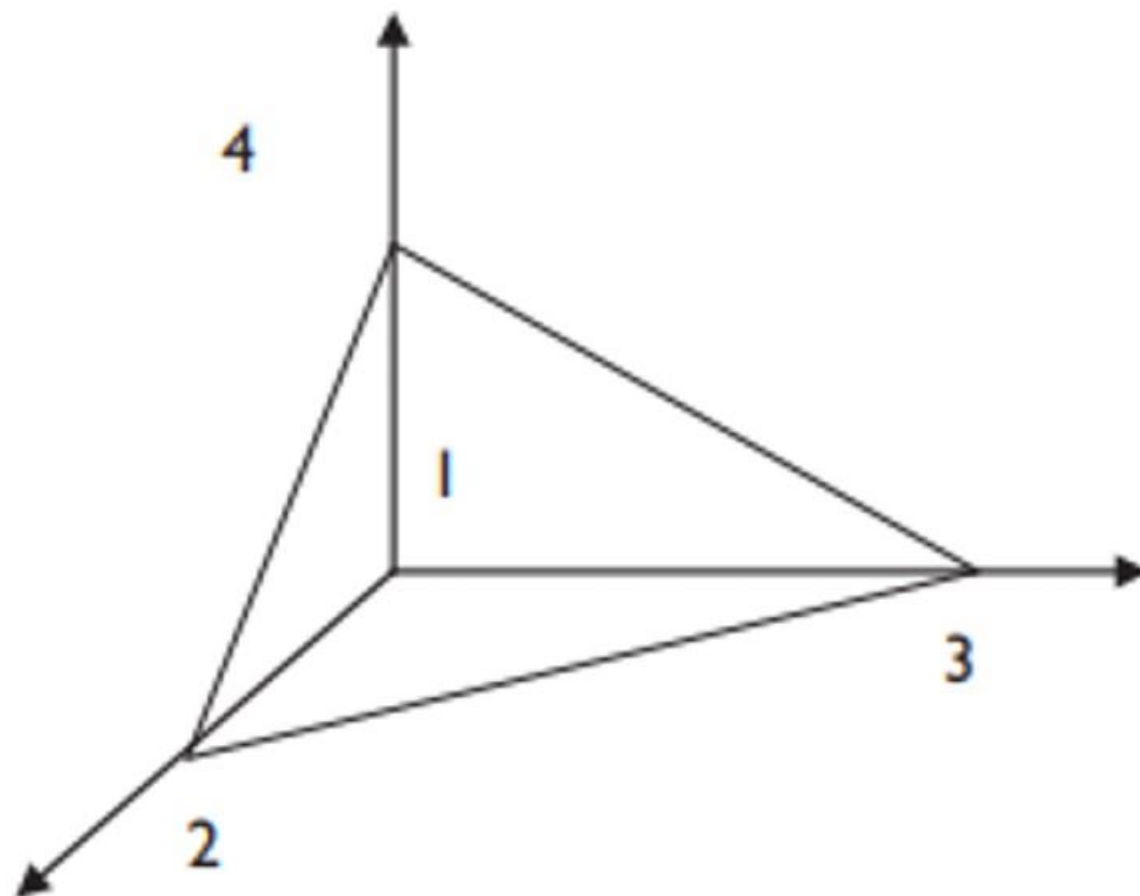
✓ in consideration of equal travel times where the new origin is positioned at receiver 1. The solution to this equation for $x_s = \frac{x_2}{2}$. This bisects the line passing through the two receivers. The source must lie in a plane perpendicular to this line and must contain this point. If the source were out of the bisecting plane, the travel times would differ contrary to the given conditions of the problem.



- ✓ **Example 9.23** Show that when two receivers detect an event at the same time, the source must lie on a perpendicular bisector of the line passing through the two receivers.
- ✓ **Solution:** Alternatively, $\delta x^2 = 2\beta^2 \left(\frac{d_1^2}{4} - \delta x^2 \right)$ $x^2 = 2\beta^2 \left(\frac{d_2^2}{4} - \delta x^2 \right)$ where the d 's are the yz distances between source and receivers. These distances must be equal to satisfy the basic, physical distance - time relationship. In essence, the event source must lie on the intersection of two spheres of equal radii emanating from the considered receivers. The intersection is a circle perpendicular to the line of receivers and the circle radius is a perpendicular bisector of the line.

Event Location

- ✓ **Example 9.24** An event occurs in 3-space and is simultaneously recorded by four receivers positioned at the vertices of a tetrahedron as shown in the sketch. Determine the event location and time.



Sketch for Example 9.24

Event Location

✓ **Example 9.24** An event occurs in 3-space and is simultaneously recorded by four receivers positioned at the vertices of a tetrahedron as shown in the sketch. Determine the event location and time.

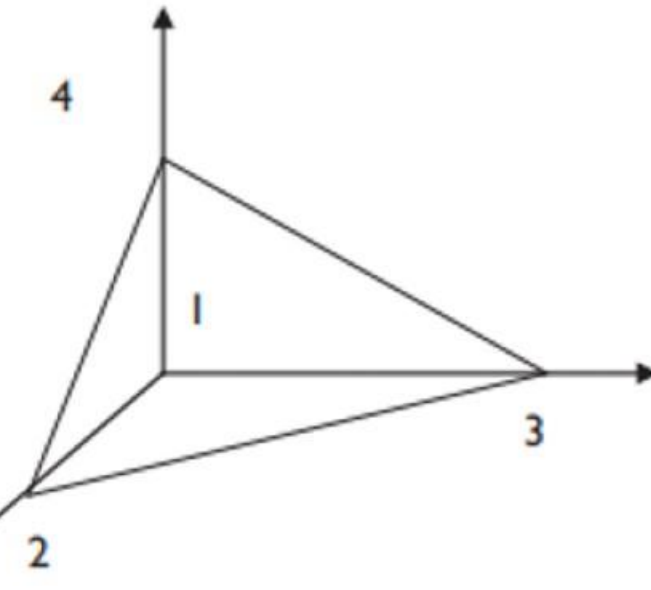
✓ **Solution:** By the guiding principle

$$\sqrt{(0 - x_s)^2 + (0 - y_s)^2 + (0 - z_s)^2} = v(t_o - t_s)$$

$$\sqrt{(x_2 - x_s)^2 + (0 - y_s)^2 + (0 - z_s)^2} = v(t_o - t_s)$$

$$\sqrt{(0 - x_s)^2 + (y_3 - y_s)^2 + (0 - z_s)^2} = v(t_o - t_s)$$

$$\sqrt{(0 - x_s)^2 + (0 - y_s)^2 + (z_4 - z_s)^2} = v(t_o - t_s)$$



Sketch for Example 9.24

✓ where the receivers sense the event at the same time t_o .

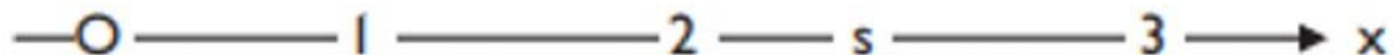
✓ The solution to this system for the source location using the first and second, then the first and third, and finally the first and fourth equations is $x_s = \frac{x_2}{2}$, $y_s = \frac{y_3}{2}$, $z_s = \frac{z_4}{2}$.

✓ Substitution of this result back into the first equation gives $t_s = t_o - \frac{1}{4}v\sqrt{(x_2)^2 + (y_3)^2 + (z_4)^2}$



Event Location

- ✓ **Example 9.25** Three receivers on a line detect an event that occurs between receivers 2 and 3. Determine the event location and time.
- ✓ **Solution:** The receivers are located on a line as shown in the sketch.



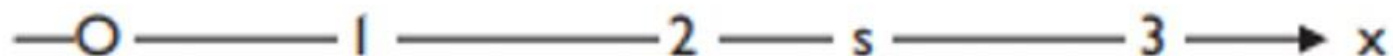
Sketch for Example 9.25



Event Location

- ✓ **Example 9.25** Three receivers on a line detect an event that occurs between receivers 2 and 3. Determine the event location and time.

- ✓ **Solution:** The receivers are located on a line as shown in the sketch.



Sketch for Example 9.25

- ✓ From the basic distance - time principle

$$(x_s - x_1) = v(t_1 - t_s) \text{ or } x_s + vt_s = x_1 + vt_1$$

$$(x_s - x_2) = v(t_2 - t_s) \text{ or } x_s + vt_s = x_2 + vt_2$$

$$(x_3 - x_s) = v(t_3 - t_s) \text{ or } x_s + vt_s = x_3 - vt_3$$

- ✓ **Comment:** In this simple case, the over determined system is readily solved by the procedure of “squaring the system”. When many receivers are considered and the event may not be boxed in by the array, the situation is not so easily resolved, although techniques are available for event location in such circumstances.



- ✓ **Question:** Two geophones are located along a tunnel at 100 ft (G1) and 1100 ft (G2) from the tunnel entrance. Longitudinal wave velocity in the adjacent rock mass is 10,000 ft/s. An event arrival is detected at G1 at 10:31.20.06 (hr, min, sec). An event arrival is also detected at G2 10:31:20.04. Determine if the event is locatable and if so, the location and time of the event.

- ✓ **Question:** Two geophones are located along a tunnel at 100 ft (G1) and 1100 ft (G2) from the tunnel entrance. Longitudinal wave velocity in the adjacent rock mass is 10,000 ft/s. An event arrival is detected at G1 at 10:31.20.06 (hr, min, sec). An event arrival is also detected at G2 10:31:20.16. Determine if the event is locatable and if so, the location and time of the event.

- ✓ **Question:** Two geophones are located along a tunnel at 10 m (G1) and 310 m (G2) from the tunnel entrance. Longitudinal wave velocity in the adjacent rock mass is 3,000 m/s. An event arrival is detected at G1 at 10:31.20.06 (hr, min, sec). An event arrival is also detected at G2 10:31:20.04. Determine if the event is locatable and if so, the location and time of the event.



- ✓ **Question:** Three geophones are located along a tunnel at 10 m (G1), 310 m (G2), and 640 m (G3) from the tunnel entrance. Longitudinal wave velocity in the adjacent rock mass is 3,000 m/s. An event arrival is detected at G1 at 10:31.20.06 (hr, min, sec). Event arrivals are also detected at G2 at 10:31:20.04 and at G3 at 10:31:15. Determine if the event is locatable and if so, best estimates of location and time of the event.



- ✓ **Question:** Three geophones are located along a tunnel at 10 m (G1), 310 m (G2), and 640 m (G3) from the tunnel entrance. Longitudinal wave velocity in the adjacent rock mass is 3,000 m/s. An event arrival is detected at G1 at 10:31.20.06 (hr, min, sec). Event arrivals are also detected at G2 at 10:31:20.16 and at G3 at 10:31:27. Determine if the event is locatable and if so, best estimates of location and time of the event.



- ✓ **Question:** Geophone pairs are installed in two drifts along a narrow, steeply dipping vein as shown in the sketch. An event occurs that may be a burst in the hanging wall. If this event is locatable, determine the source location (x_s, y_s) and time of the event (t_s). Assume the origin of coordinates is at G1 (0,0) and the coordinates of the other geophones are: G2(0,200), G3(200, 150), G4(0,150), so the geophones are at the corners of a plane rectangle. Units are ft. Relative to an arbitrary zero of time, arrival times in milliseconds are: G1(20.0), G2(32.36), G3(35.50) and G4(25.81). Wave speed for first arrivals (P-waves) is 7,071 ft/s. Comment: The location requires extreme accuracy in event times. The problem was setup with the source at ($x=50.0, y= 50.0, t= 0.010$).

Thank You

Queries??

Email: ashokmin@iitism.ac.in
Mobile: 8603401161